

JEE MAIN REVISION MASTER: WAVES

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 9 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Wave Velocity, Standing Waves & Organ Pipes	5 PYQs	Acoustics & Waves
Moderate	Beats & Doppler Effect for Sound	4 PYQs	Wave Phenomena

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Wave Velocity, Standing Waves & Organ Pipes	$f_{\text{closed}} = \frac{v}{4L}, \frac{3v}{4L}, \frac{5v}{4L}; \quad f_{\text{open}} = \frac{v}{2L}, \frac{2v}{2L}, \frac{3v}{2L}; \quad v_{\text{string}} = \sqrt{\frac{T}{\mu}}$
Beats & Doppler Effect for Sound	$f_b = f_1 - f_2 , \quad f' = f \left(\frac{v - v_o}{v - v_s} \right) \quad (\text{towards each other: } f' > f)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Wave Velocity, Standing Waves & Organ Pipes (Moderate Yield — 5 PYQs)

Core Principle: Newton-Laplace formula: $v = \sqrt{(\gamma P / \rho)}$. Closed organ pipe produces only ODD harmonics: $f_n = (2n-1)(v / 4L)$ (ratio 1:3:5). Open organ pipe produces ALL harmonics: $f_n = n(v / 2L)$ (ratio 1:2:3:4). Fundamental frequency of open pipe is twice that of closed pipe of same length.

■■ **Caution / JEE Pitfall:** In organ pipes, remember: Third harmonic = First overtone in a closed pipe, but Second harmonic = First overtone in an open pipe!

■ **Speed Trick:** Speed of sound in a gas is INDEPENDENT of pressure changes at constant temperature because (P / ρ) remains constant.

Representative Source PYQs:

- [2025 Shift] Q#2 (MCQ): "The angle between the particle velocity and wave velocity in a transverse wave is [except when the particle passes through the mean position..." [Ans: N/A]
- [2021 Shift] Q#1 (MCQ): "To propagate both longitudinal and transverse waves, a material must have..." [Ans: N/A]
- [2019 Shift] Q#3 (MCQ): "The equation of a stationary wave is $y=2\sin((\pi x / 15))\cos(48\pi t)$. The distance between a node and its next antinode is..." [Ans: N/A]

3.2 Beats & Doppler Effect for Sound (Moderate Yield — 4 PYQs)

Core Principle: Beat frequency $f_b = |f_1 - f_2|$ produced by superposition of two waves of slightly different frequencies. Loading a tuning fork with wax decreases its frequency; filing a fork increases its frequency. Doppler effect: $f' = f \left(\frac{v \pm v_o}{v \mp v_s} \right)$.

■ **Caution / JEE Pitfall:** When source moves TOWARDS stationary observer, use minus sign in denominator: $f' = f \left(\frac{v}{v - v_s} \right) > f$.

■ **Speed Trick:** If a tuning fork A (frequency f_A) gives 4 beats/s with B, $f_B = f_A \pm 4$. If loading A with wax decreases beat count, $f_A > f_B$.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "A source of frequency ν gives 6 beats/second when sounded with a source of frequency 200 Hz. The second Harmonic of frequency 2ν ..." **[Ans: B]**
- **[2024 Shift] Q#3 (MCQ):** "A motor-cyclist moving towards a huge cliff with a speed of 18 km h^{-1} , blows a horn of source frequency 325 Hz. If the speed of the sou..." **[Ans: N/A]**
- **[2024 Shift] Q#4 (MCQ):** "A galaxy is moving away from the Earth so that a spectral line at 600 nm is observed at 601 nm. Then, the speed of the galaxy with respect ..." **[Ans: A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A source of frequency ν gives 6 beats/second when sounded with a source of frequency 200 Hz. The second Harmonic of frequency 2ν of the source gives 8 beats/second when sounded with a source of frequency 420 Hz. The value of ν is

- (A) 205 Hz
- (B) 206 Hz **[Correct]**
- (C) 195 Hz
- (D) 210 Hz

Correct Answer: B

Solution Analysis: For beats, the number of beats per second is equal to the difference in the two frequencies. Given that the source of frequency ν gives 6 beats/second with 200Hz : $|\nu - 200| = 6$ So, $\nu = 200 + 6 = 206 \text{ Hz}$ or $\nu = 200 - 6 = 194 \text{ Hz}$ Now the second harmonic of the source has frequency 2ν . It gives 8 beats/second with 420Hz, so $|2\nu - 420| = 8$ This gives $2\nu = 420 + 8 = 428$ or $2\nu = 420 - 8 = 412$ So, $\nu = 214 \text{ Hz}$ or $\nu = 206 \text{ Hz}$ The common value from both conditions is $\nu = 206 \text{ Hz}$ So the correct option is Option B: 206 Hz.

Benchmark Problem 2: KCET 2025 (Q#2)

The angle between the particle velocity and wave velocity in a transverse wave is [except when the particle passes through the mean position]

- (A) $(\pi / 4)$ radian
- (B) $(\pi / 2)$ radian
- (C) π radian
- (D) Zero radian

Correct Answer: None

Solution Analysis: In a transverse wave, particles oscillate in a direction perpendicular to the wave's direction of propagation. Therefore, the angle between the particle velocity and wave velocity is $(\pi / 2)$ radians, except when the particle crosses the mean position.